DATA VISUALIZATION DESIGN PROJECT · MAY 2026 · GROUP 2

A Visual Study of E-Commerce Orders, Delivery & Customer Satisfaction

What drives a happy customer vs. a one-star review — and where the marketplace should invest to grow without breaking the experience.

Olist Brazilian E-Commerce dataset + Marketing Funnel dataset · 99,441 orders · Sept 2016 – Aug 2018

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Delivery Performance

Ashwanth
Product & Pricing

Kannan
Seller Performance

Anushka
Regional Logistics

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Seller Behaviour & Investment

THE PROBLEM

Growth and satisfaction pull in opposite directions

Onboard sellers aggressively and gross sales rise — but late deliveries, poorly-rated products, and bad reviews start driving customers away. Crack down on every slow seller and quality improves — but the catalogue shrinks and growth stalls. Every decision leadership makes trades growth against satisfaction, and right now those calls are made on gut feel.

12.3%
of orders end in a 1–2 star review

96,478
delivered orders analysed

3,095
active sellers on the platform

THE GUIDING QUESTION

Which product categories, seller behaviours, regions, and delivery patterns predict a happy customer vs. a one-star review — and where should the company invest to grow the marketplace without breaking the customer experience?

OUR APPROACH

Five members, five viewpoints, one shared definition of a bad experience

A “bad experience” is a review of 1–2 stars out of 5. Reviews are U-shaped (57% five-star, 12% one-star), so every analysis in this deck uses the negative rate, not the average score — an average of ~4.1 sits in a gap almost no customer is actually in.

DELIVERY	**Yash Arabhavi**	Delay ⟶ reviews · Slow vs. broken promise
	Anushka	Regional performance · Worst routes
SELLERS	**Kannan**	Damage concentration · Repeat offenders
	Sahib Randhawa	Behavioural drivers · Where to invest
PRODUCT	**Ashwanth**	Beyond delivery · Category quadrant

Missing the promised date is a cliff, not a slope

9.2% → 62.3%

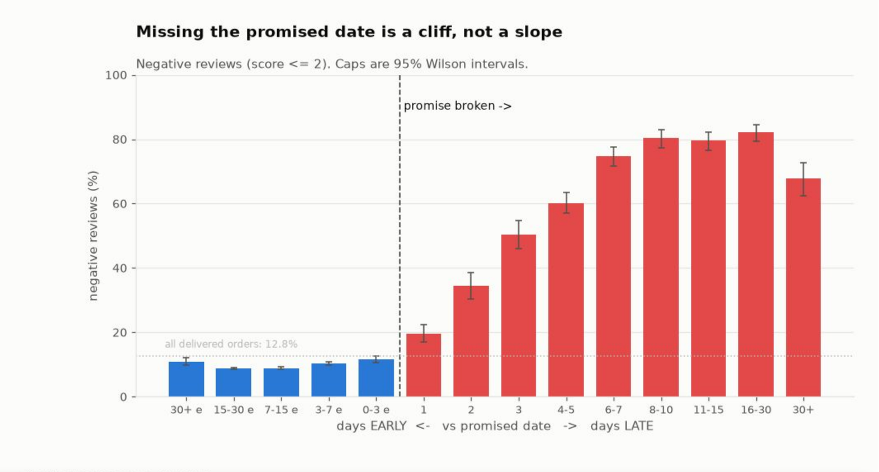
On-time orders are 9.2% negative; late orders are 62.3% negative — 6.8× worse.

6.7% of orders, 32.5% of bad reviews

Late deliveries are a small share of volume but drive a third of all negative reviews.

Flat before, cliff after

Across 60 days of being early, the negative rate barely moves (8.9–11.6%). One day late: 19.6%. Three days late: 50.4%.



CAVEAT

2,849 reviewed orders never arrived at all (78% negative) — they're excluded from this delay analysis since an order with no delivery date has no delay to measure.

It's the broken promise, not the wait, that angers customers

21.2% vs 25.4%

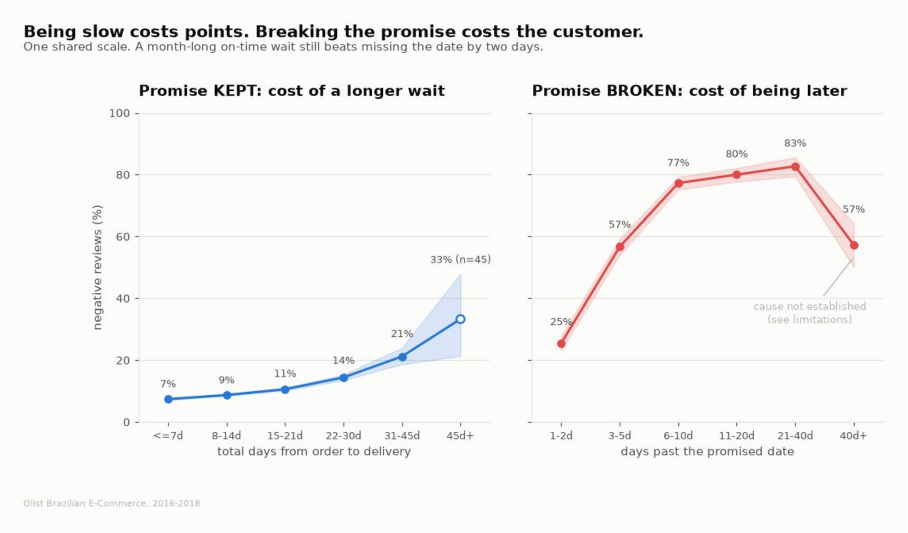
A month-long on-time wait (21.2% negative) still beats being just 1–2 days late (25.4% negative).

74.5% vs 0.3%

Share of customers surveyed before their parcel even arrived, for late vs. on-time orders — late customers often rate a parcel they don't have yet.

RECOMMENDATION

People aren't rating how long they waited — they're rating the gap between what they were told and what happened. The cheapest fix is a more honest promised date, not faster logistics: the estimate is platform-controlled, so it's the one lever here that needs no investment in couriers.



Cost of waiting when the promise holds (left) vs. when it's broken (right) — same shared scale

Do certain regions experience systematically worse delivery performance and satisfaction?

ANALYSIS PENDING — ANUSHKA TO COMPLETE (Q1)

KEY FINDING

CHART

RECOMMENDATION

Suggested starting point: late-delivery rate & negative-review rate by customer_state, with confidence intervals; use hollow/faded bars where a state has under ~100 orders (see Yash's Fig. 1 convention).

Which seller-state → customer-state routes perform worst?

ANALYSIS PENDING — ANUSHKA TO COMPLETE (Q2)

KEY FINDING

CHART

RECOMMENDATION

Suggested starting point: a seller_state × customer_state heatmap of the negative-review rate, with a minimum-order threshold per route; check whether distance or interstate shipping explains the worst routes, or whether it's route-specific.

The worst 5% of sellers cause 58% of bad reviews — but they also handle 53% of orders

+5pp, not +58pp

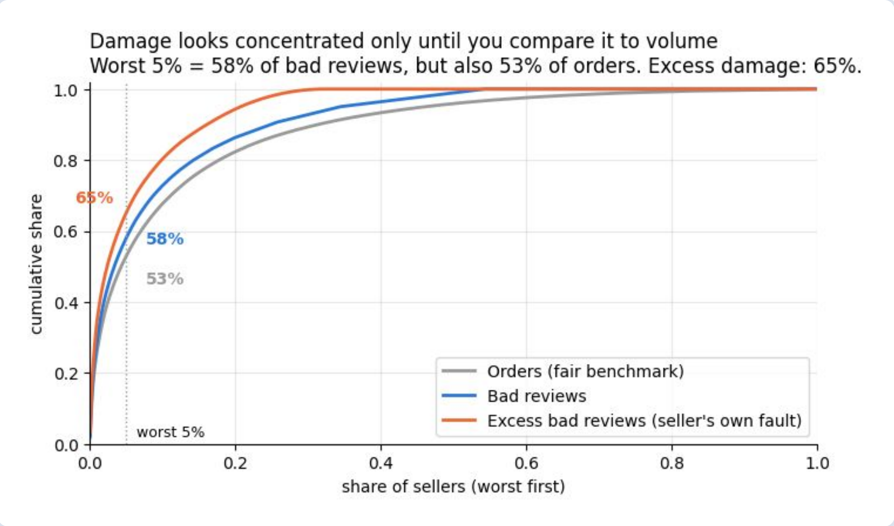
The worst 5% by bad-review count handle 53% of all orders — so their real excess over a fair share is only +5 percentage points.

65% of avoidable damage

Once category and delivery effects are removed, the worst 5% of sellers own 65% of the bad reviews their record can't explain.

WHY THIS MATTERS

The headline “worst 5% cause 58% of bad reviews” is misleading on its own — it mostly rediscovers seller size. Acting on raw counts would target the platform's biggest partners. Excess damage is the number to act on.



Cumulative share of orders vs. bad reviews vs. excess (avoidable) damage, sellers ranked worst-first

Yes — 77 sellers are repeatedly worse than their category and delivery record predicts

2.4× worse, even on time

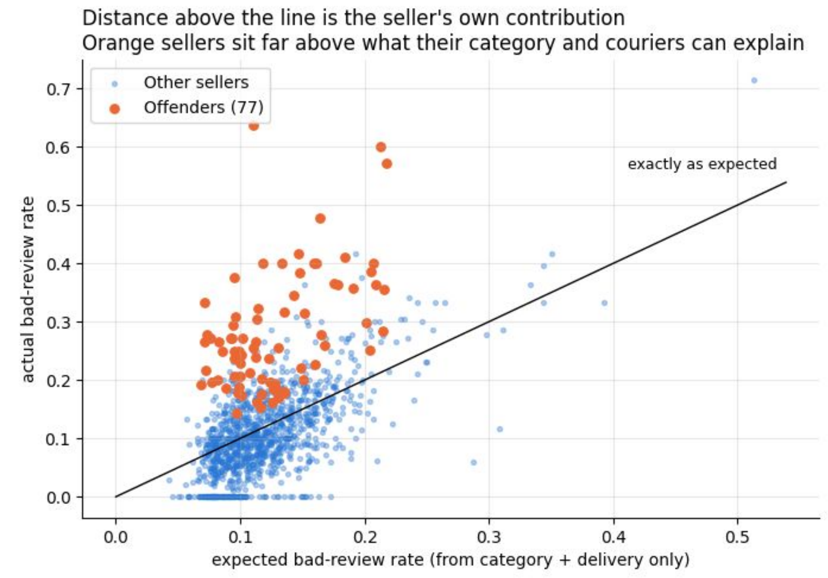
Removing the logistics excuse entirely, these 77 “offenders” draw an 18.4% bad-review rate vs. 7.6% for everyone else.

20× more than chance

33 sellers sit 3+ standard deviations above their own expectation — pure random noise would produce fewer than 2.

11% of GMV, 53% of avoidable damage

A small, identifiable group responsible for the majority of the damage that category and delivery can't explain.



Every seller's actual bad-review rate vs. what its category + delivery record predicts

Coach the 77, don't delist them

Coach to median: −1.30pp, 0% GMV lost

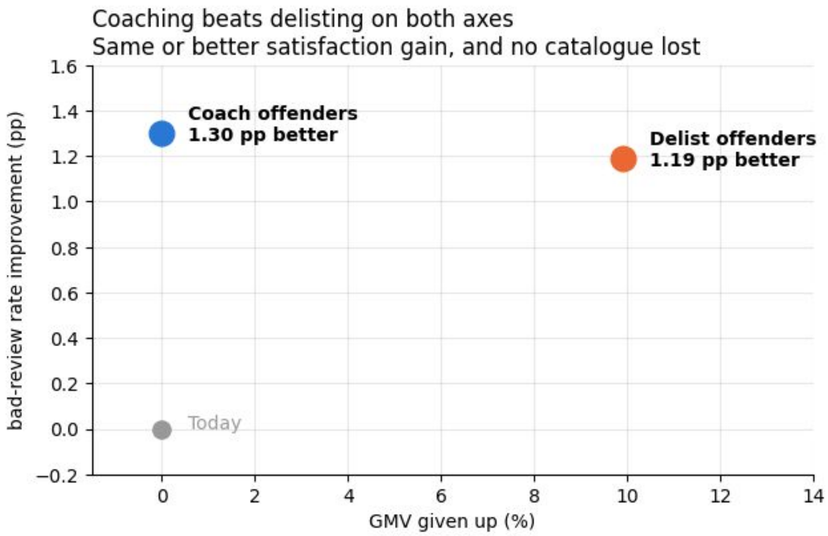
Bringing the 77 offenders to the median seller's performance improves the platform bad-rate with no revenue given up.

Delist instead: −1.19pp, −9.9% GMV

A smaller satisfaction gain, and it costs a tenth of platform revenue — delisting removes the ~78% of an offender's orders customers were happy with too.

SPLIT BY FAILURE MODE

26 of the 77 ship fast (3.7% late — better than average) but the item still disappoints. No shipping SLA fixes this group; they need a listing and quality review instead. Delisting should be a last resort, not the opening move.



Platform bad-review-rate improvement vs. GMV given up, for each intervention

Handling time is the dominant driver — volume and freight pricing are not reliable predictors

r = +0.24, p < 0.0001

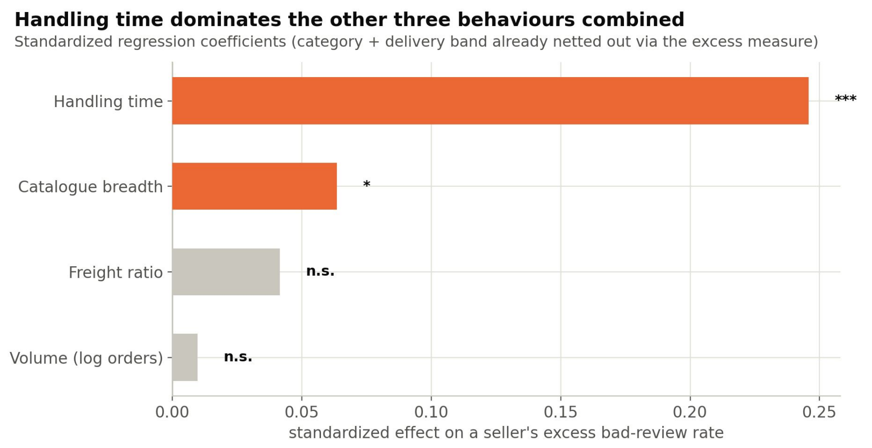
Handling time is the only behaviour with a strong, highly significant correlation to a seller's excess bad-review rate.

Volume & freight: not significant

Order volume (r=+0.02) and freight ratio (r=+0.04) show no reliable link to excess bad reviews, controls held.

Robust to controls

The effect survives a multiple regression (β=0.0079, p<0.0001) and holds on-time orders only.



WHY THIS MATTERS

Catalogue breadth has a small, marginal effect (p=0.035) but is dwarfed by handling time — seller-quality investment should prioritize handling-time coaching over volume caps or freight rules.

Fund the zero-cost delivery-promise fix first, then sequence seller coaching by handling time

3.56pp vs 1.30pp

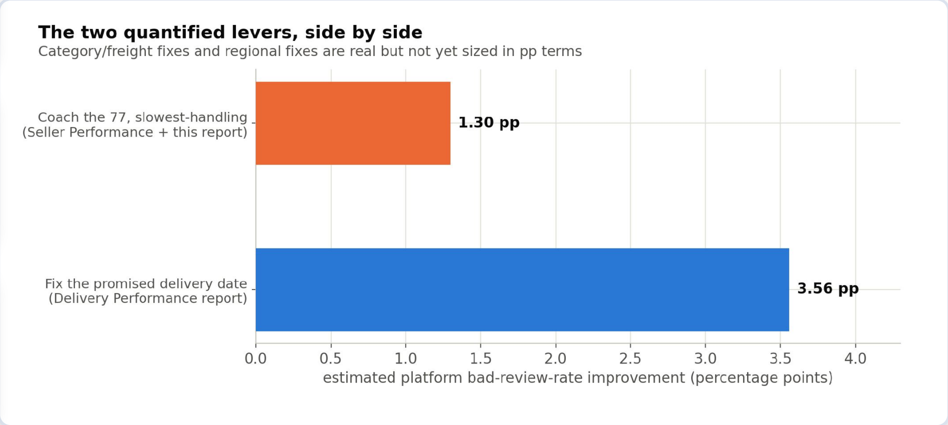
The delivery-promise fix outweighs seller coaching roughly 2.7 to 1 — both cost nothing in lost GMV.

77 sellers, coached not delisted

Coaching the 77 flagged sellers to the median cuts bad reviews with 0% GMV given up — sequence them by handling time.

Two levers not yet sized

Category/freight fixes (58% of bad reviews from 10 categories) and regional fixes are real but not yet sized in pp terms.



RECOMMENDATION

Fix delivery honesty first (~3.6pp, zero cost), then sequence seller coaching by handling time (1.30pp) — extend the other two findings to a matching pp estimate before ranking them.

Beyond delivery: freight fairness and order complexity move the score

No. of items: strongest driver

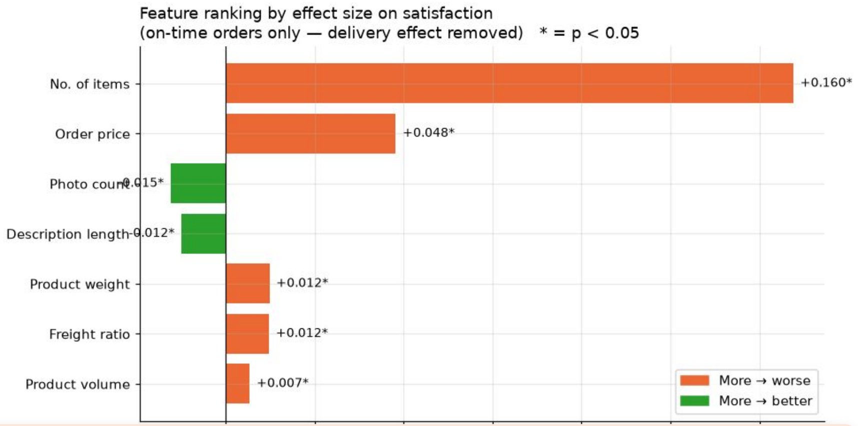
Multi-item orders have more points of failure — 4+ items: 32% negative vs. 8% for single-item orders.

Freight ratio: strongest price signal

Orders where freight exceeds ~50% of item price see nearly 2× the bad-review rate — customers feel the shipping cost is unfair.

Photos & descriptions help

More photos (up to ~4) and longer descriptions modestly reduce bad reviews by setting accurate expectations.



RECOMMENDATION

Two low-cost fixes: (1) flag single-photo listings and require sellers to add more images; (2) flag orders where freight exceeds 40–50% of item value as a price-transparency risk before purchase.

Ten categories carry over half of revenue — with worse-than-average satisfaction

High Rev + Poor Sat (fix urgently)

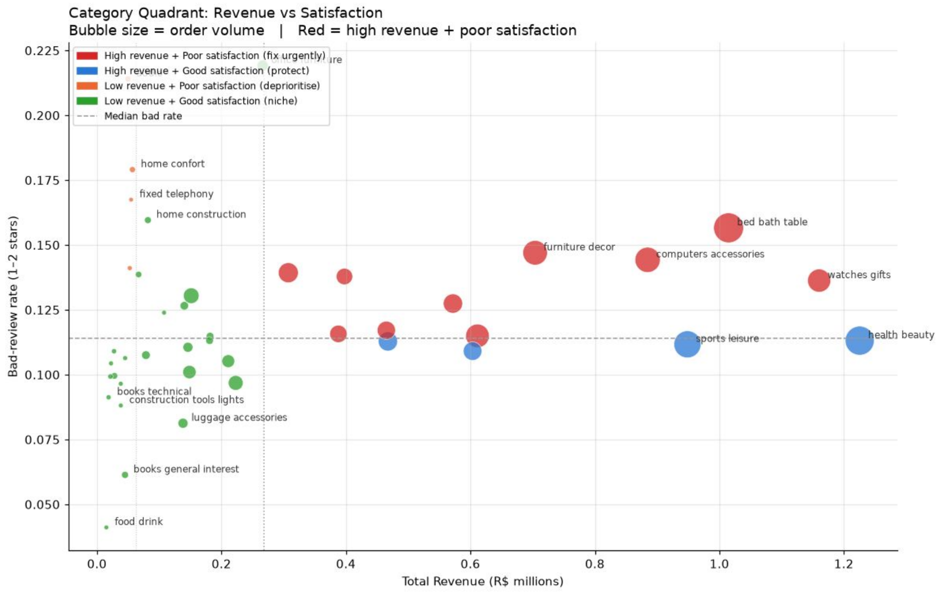
10 categories = 52.5% of revenue, 58.2% of bad reviews. Watches & gifts, bed/bath/table, computers accessories, furniture décor...

High Rev + Good Sat (protect)

Health & beauty, sports & leisure, toys, cool stuff — the platform's engine. Monitor closely as they scale.

RECOMMENDATION

Apply the Q1 fixes (photo counts, freight transparency, seller coaching) to the 10 red categories first — that's where quality investment has the highest revenue-at-risk payoff.



Revenue vs. satisfaction by category — bubble size = order volume

LIMITATIONS & NEXT STEPS

Further Research Analysis

- **Patterns, not proof**

  Every finding here is an association established through careful controls (delivery, category) — not a causal model.

- **Silent churn is invisible**

  Orders that never arrive are 2.9% of reviewed orders but 15.3% of all bad reviews, and reviews only come from customers who respond to the survey at all.

- **Two questions still open**

  Regional/route performance (Anushka) and seller-behaviour drivers plus the investment plan (Sahib) — see the placeholder slides for suggested approaches.

- **Single marketplace, two years**

  Nothing here has been validated outside this dataset (Sept 2016 – Aug 2018, one Brazilian marketplace).

Thank you — Group 2